

Capacity is an interval: the traffic rule sets the floor, the network sets the ceiling

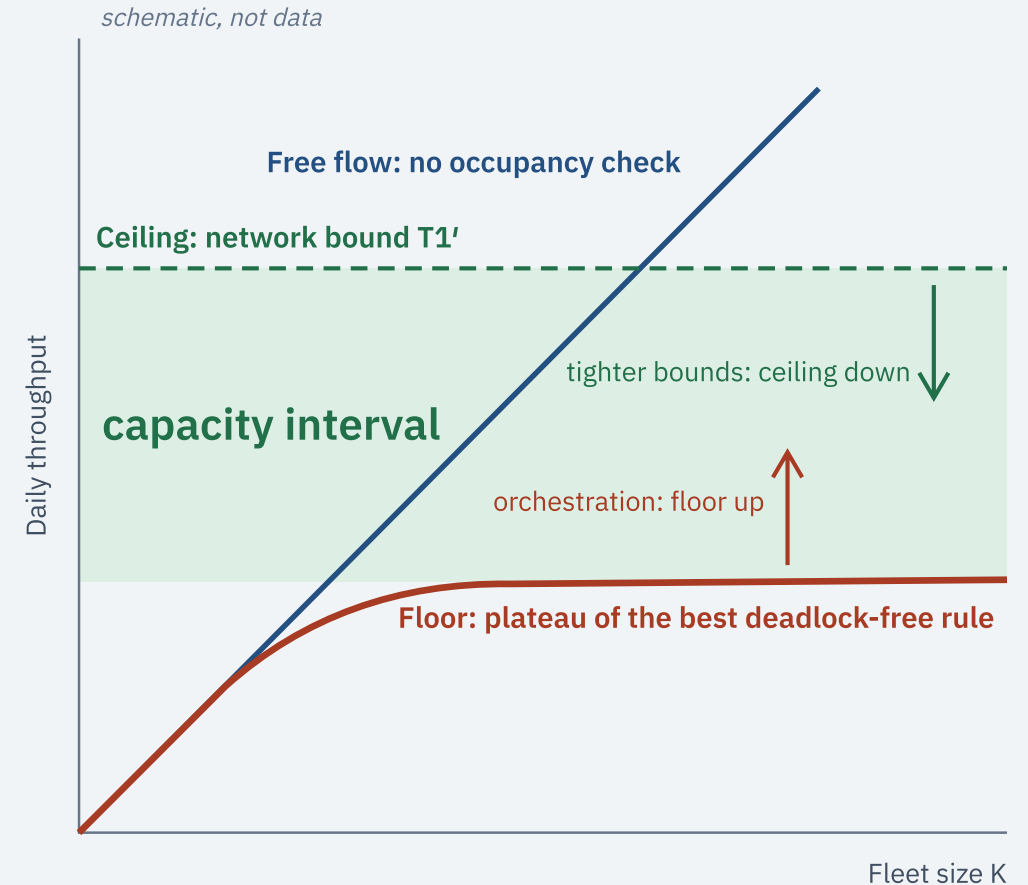
The flag experiment on two real yards: the capacity interval, the fleet-throughput trapezoid, the network-bound regime and the free-flow error (C1, C2, H1, H3)

When transporters own the road: capacity bounds and operating regimes in shipyard block logistics (working title)

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Data: thesis-core commit a882742 (T17–T23, all reviewed); literature anchors from T24



The sub-study in one minute

1

Question

When loaded vehicles own the road, how much can block transport deliver, when do more vehicles stop helping, and how wrong is free flow?

2

Method

Two real networks, Okpo and Yantai; four traffic rules, two granularities; fleets up to 150, 10 seeds each; an LP gives the bound T1'; vehicles needed at whole-yard volume, 16-h and 24-h days.

3

Results so far

Capacity intervals: Okpo [627, 1,720], Yantai [455, 1,252], the floor from the best deadlock-free rule at 36% of the ceiling on both. At Okpo's own 339 a day reservation needs 27 vehicles (16 h), below $K^* 50\text{--}60$; at the peak of 678, 117.

4

Next

T27 tests a less conservative deadlock-free rule, re-checking the floor and H2's fall; after it the author settles H1's main caliber (16 h or 24 h, mean or peak); then T25 (retrofits, exchange rate) and T26 (floating docks, holding area).

Capacity is not one number but an interval: the traffic rule sets its floor, the network its ceiling.

This sub-study answers four items of the research manifesto: C1, C2, H1 and H3

Manifesto wording (v1, 9 Oct 2026)		What this sub-study delivers	
C1	Throughput $\leq \min\{\text{fleet bound } K \cdot H / \bar{c}, \text{network bound}\}$. The network bound is set by the blocking-time structure and can be computed by linear programming. Attainable capacity lies within the interval [best deadlock-free rule, network bound]		Intervals for both yards, both granularities and the erection mix, the floor from the best tested deadlock-free rule; system T1' and route-free T1'' firm up the ceiling
C2	Throughput first rises linearly with fleet size, then enters a plateau set by the network; when waiting occupies the road, it falls beyond a jamming threshold. The free-flow model sees neither of the last two parts		First two parts: both yards' flag curves; the fall belongs to H2: teleport-free, deadlock-free safe release rises then falls, re-checked by T27
H1	At a reasonable daily task volume and a realistic fleet size, the system is in the network-bound regime, at least at erection peaks or near the dock mouth		K^* ; integer vehicles needed (16 h and 24 h) judged against the K^* range; the density criterion; the work-zone chart; dock-mouth thresholds by fleet size
H3	The free-flow model's underestimate of completion time and of vehicles needed grows superlinearly with vehicle density		The exponent α of the service-time gap $\Delta(K)$; the ratio of vehicles needed at whole-yard volume (integer K , 16 h and 24 h)

The flag experiment: two real yard layouts, four traffic rules, two resource granularities, one task stream

Free flow	No occupancy check; vehicles never interact (the field's default)
Whole-route reservation	Books the whole route before a loaded departure, waits off the road; deadlock-free by construction. The best tested such rule: the interval's floor
Safe release	Segment requests granted after a Banker-type safety check; no teleport, never deadlocks; exact but claims conservatively (new in T22)
Segment request	One segment at a time; on deadlock a vehicle leaves the network for 900 s Reference rule (teleport clearing): supports no C or H on its own

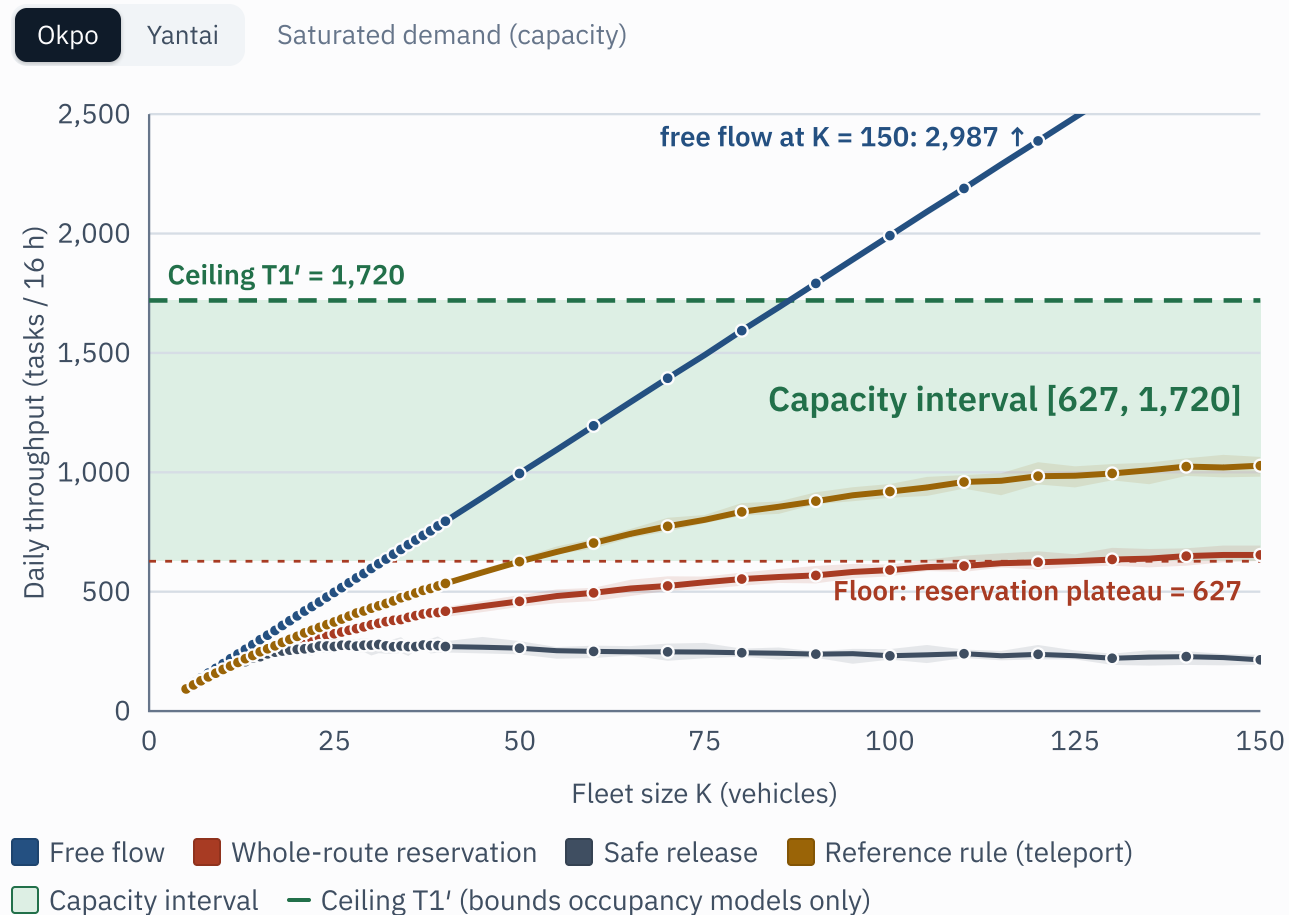
- **Demand:** saturated to measure capacity (K = 5–150, Yantai 1–150, 10 seeds, 4 days); finite whole-yard volumes of 300–1,200 a day plus Okpo's own 339 and peak 678, on 16-h and 24-h days; 97 / 194 is a four-transporter subset, for comparison only

	OKPO	YANTAI
Roads (count / km)	721 / 40.3	140 / 8.6
Junction resources	334	58
Stops	421	140
Own daily volume: mean / peak	339 / 678	23 / 46
Where T1' binds (seeds)	dock road 161 5/10, road 010-6 4/10	the two doors of building 014 9/10

- **Ceiling T1':** an occupancy LP with width bin-packing rows (Fekete–Schepers), exact rationals; checked case by case, none exceeds it (17,760 runs in T21, 40,157 more in T22)
- Maps and granularity:** Hanwha Ocean Okpo derived map rev 1825 (network as rev 1824 in T18), CIMC Raffles Yantai rev 396 (identical to rev 395), traced by the author on public imagery; neither imagery nor geometry is shown. Whole road: each road one resource; segmented: dock roads blocked at their stops (one rule for both yards); after the author's imagery check segmented is closer to Okpo's practice, and both are reported. From T27's step zero, maps come from 20_数据/船厂地图 (Okpo rev 1833).

Sources: thesis-core a882742, docs/T17_flag_report.md §1–2, T18 and T20–T22 reports (T22 §2.3, §8); config.json of results/a0ab16fceb94, 49d2383872e7, 656e2da75368 (Okpo) and b82aa19b22af, 930e637e0658 (Yantai). Sources of speeds, platform size and handling times: T17 report §2; the map switch: 00_总控/tasks.md, row T27.

The capacity interval [best deadlock-free rule, T1']: Okpo [627, 1,720], Yantai [455, 1,252]; on both the floor is only 36% of the ceiling



[627, 1,720]

Capacity interval (a day): the floor is the reservation plateau over K = 100–150, the ceiling T1' (mean of 10 seeds, 1,436–1,948 across seeds)

36%

The floor reaches only this share of the ceiling: the gap is the traffic rule, which orchestration aims to close

229_{279 @ 31}

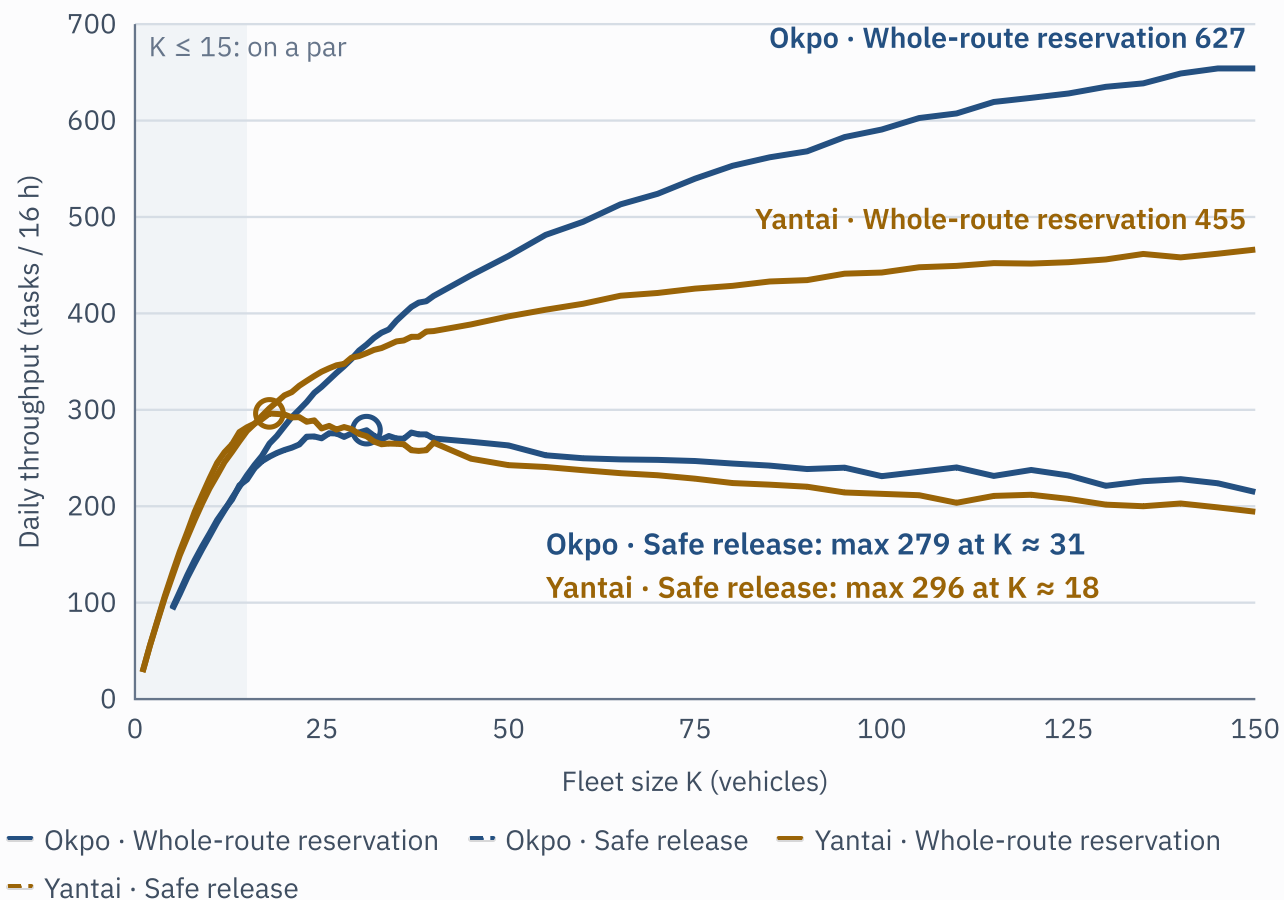
Safe-release plateau (no teleport, never deadlocks); maximum 279 at K ≈ 31, then falling, below reservation (next slide)

984_{57%}

Reference-rule plateau (teleport clearing), for reference only

Sources: thesis-core a882742; Okpo results/a0ab16fceb94, 49d2383872e7 (free flow, reservation, reference rule) and 656e2da75368 (safe release), Yantai b82aa19b22af and 930e637e0658; the interval as in T22 §4 and t22_interval.csv. Plateaus are means over K = 100–150; curves are means of 10 seeds, shading min–max. Conditions: saturated demand, each road one resource; Okpo rev 1825, Yantai rev 396. T1' bounds only the occupancy models; free flow checks no occupancy and may cross it.

The floor is reservation: safe release is exact and never deadlocks, but conservative, and lower from $K = 20$



Exact, never deadlocks

Its decisions match a naive reduction on the same input in all 860,291 checks; not one deadlock in any batch, and no vehicle leaves the network.

Conservative claims

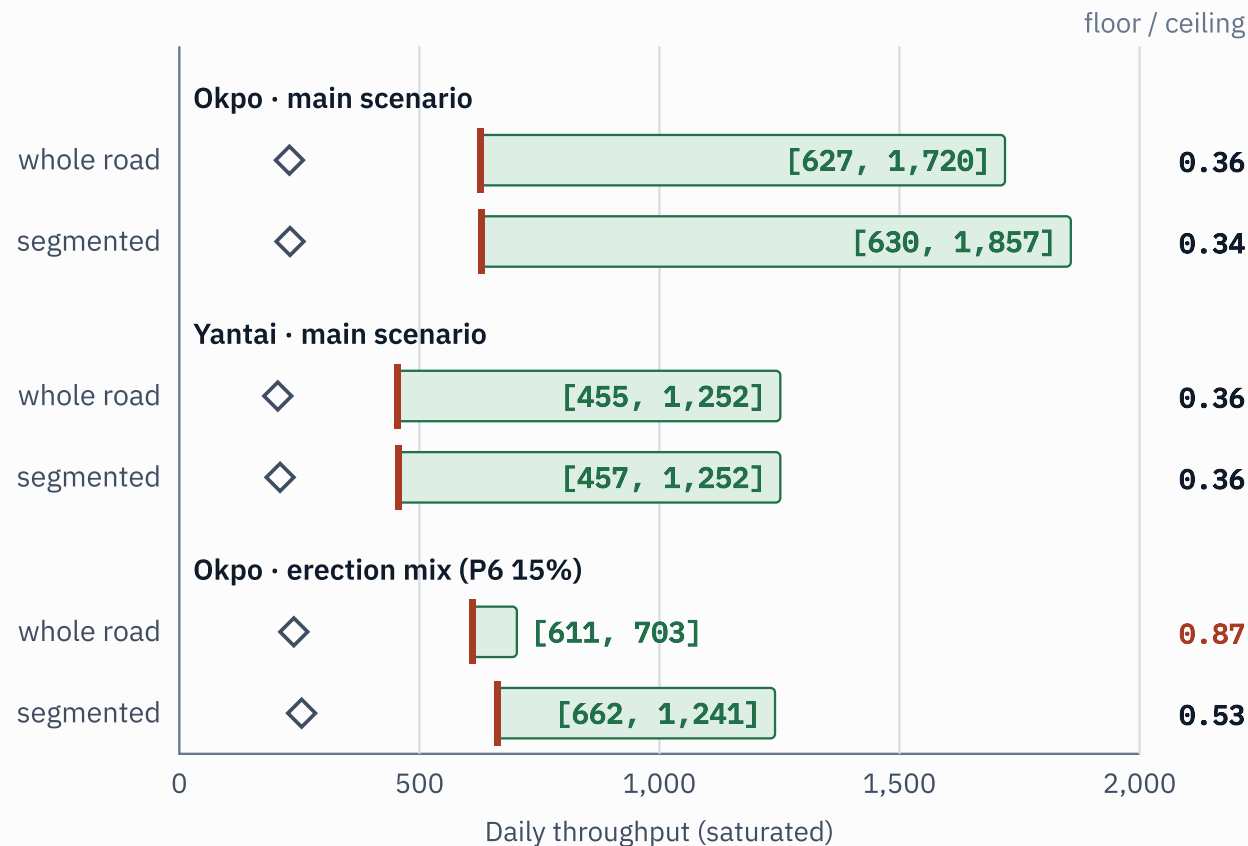
It claims each vehicle's whole remaining task and ignores timing: about 2/3 of its waits start from a rejection by the criterion; in the ring6 case three vehicles following each other round a ring, never meeting, are still refused.

Result and next step

On a par with reservation at $K \leq 15$, below it from $K = 20$, so both floors use reservation. T27 tests less conservative rules: Banker plus a step-wise feasibility search, or claims cut at the next safe holding point (T5).

Sources: thesis-core a882742, docs/T22_report.md §2.1, §4, §5; t22_naive.json (860,291 checks); test test_safe_conservative_ring6; curves from results/656e2da75368, 930e637e0658 (safe release) and the T18 and T20 batches (reservation). Conditions: saturated demand, each road one resource; Okpo rev 1825, Yantai rev 396. Safe release mirrors signalmen granting passage in the field (KED Global, 26 Mar 2024); the Banker-type criterion is borrowed from deadlock avoidance in operating systems (Habermann 1969, to be checked).

Segmenting the dock road: main scenario, ceiling only (+8%); dock-bound erection mix, both ends (+76% / +8.3%)



Capacity interval Floor: reservation plateau safe-release plateau

C4 side: the bound binds, road changes pay

Okpo erection mix, whole road: floor / ceiling = 0.87 (0.86 on the completed mix, 0.92 for the reference rule). Segmenting dock road 161 lifts the ceiling by 76%: control granularity is a first-order item on the retrofit menu.

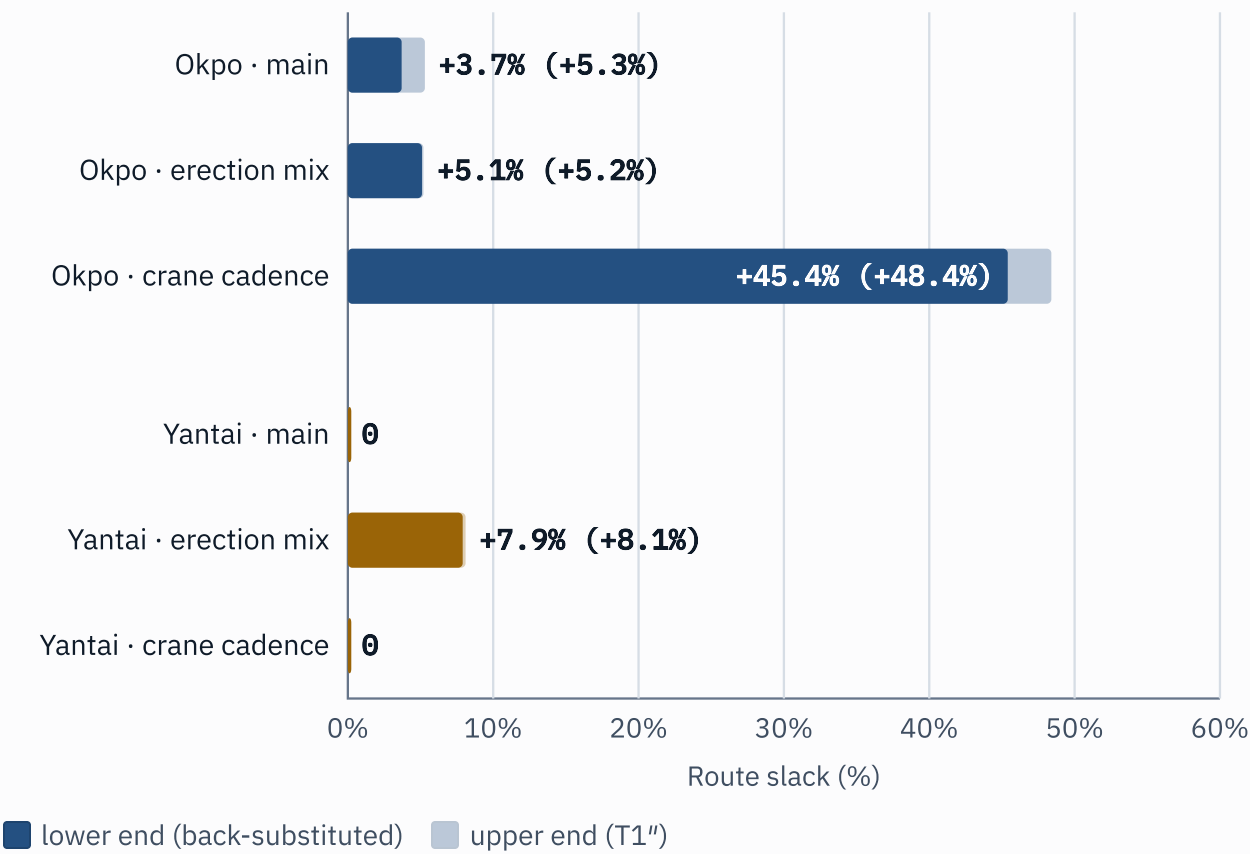
C3 side: the plateau is far below the bound, orchestrate first

Both main scenarios 0.34–0.36 (on the completed mix: reservation 36–45%, reference rule 50–57%); the same erection mix falls to 0.53 once segmented: change the granularity and the bottleneck moves from the road to the rule.

These are two ready examples of the C3 / C4 boundary that manifesto C4 states; T25 turns it into a decision rule with retrofit simulations and the road–vehicle exchange-rate table.

Sources: thesis-core a882742, docs/T22_report.md §3.5, §4 and t22_interval.csv (plateaus are means over $K = 100$ –150, recomputed and checked by make_data.py from each batch's curves.csv); the completed-mix caliber in docs/T23_report.md §4.4 and results/97d667d16273/platform.csv. Segmenting rule: roads longer than 200 m with roadside stops get block boundaries 25 m before and after each stop (one rule for both yards, a scenario setting). Conditions: saturated demand; Okpo rev 1825, Yantai rev 396.

A firmer ceiling (T23): noise-free system T1' and route-free T1"; the route slack depends on what binds

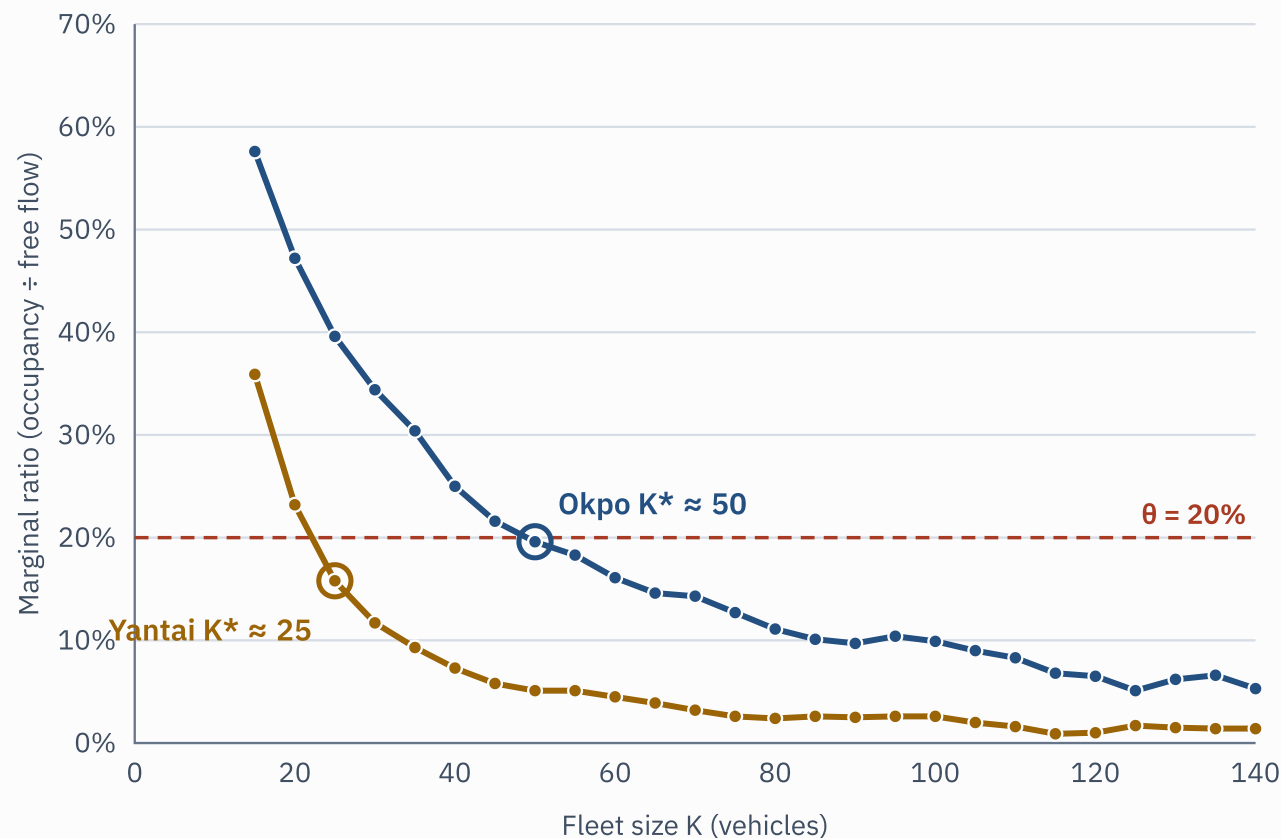


Yard · scenario	System T1'	10 seeds	T1''
Okpo · main	1,736	1,436–1,948	1,828
Okpo · erection mix	697	624–783	733
Okpo · crane cadence	1,794	1,661–1,969	2,662
Yantai · main	1,298	1,157–1,318	1,298
Yantai · erection mix	1,110	971–1,212	1,200
Yantai · crane cadence	1,225	1,111–1,240	1,225

- **Entrances:** zero slack by construction (the binding entrance has a single incoming arc)
- **Dock stopping roads:** little through traffic (under 1% on road 161); the slack comes from entering at the other end, +3.7% to 7.9%
- **Through corridors:** under crane cadence 97% of the flow on road 010-6 is passing through; routing round it gives +45.4%

Sources: thesis-core a882742, docs/T23_report.md §0, §1.2, §1.4, §2.2, §2.6 and docs/T23_route_free_proof.md; system.csv, route_free.csv and readout.csv in results/97d667d16273 and e7f94238b96d. Route slack is an interval: lower end = back-substituted feasible value – system T1', upper end = T1'' – system T1'; the independent review found no counterexample to T1''. System T1' bounds steady-state operation; chapter 3 quotes the system value and the case-by-case check keeps per-seed T1'; a draft Jensen proposition: the harmonic mean of sampled T1' does not exceed the system value in expectation. The author rewrites the draft proofs of T1' and T1''.

H1 criterion $K^*(20\%)$: from $K \approx 50$ on Okpo and $K \approx 25$ on Yantai, one more vehicle adds less than 20% of its free-flow gain



■ Okpo ■ Yantai — $\theta = 20\%$

$K \approx 50$

Okpo: network-bound from here (range 50–60)

$K \approx 25$

Yantai: a small network starts much earlier

58% → 34%

Okpo, $K = 15 \rightarrow 30$: one more vehicle's gain as a share of free flow; usual fleet sizes are already in the transition

Manifesto criterion: $K^*(20\%)$ is where the marginal throughput of one more vehicle first falls below 20% of free flow and stays below thereafter.

Sources: thesis-core a882742, t18_summary.json (Okpo) and t20_summary.json (Yantai rev 395, network identical to rev 396); reproduced with the same definition in T22 §4 (Okpo 50 / 100, Yantai 25 / 35). Marginal ratio = occupancy gain ÷ free-flow gain over a 20-vehicle window centred on K ; the range is read from the ratio's 95% interval; at $\theta = 10\%$: Okpo 100, Yantai 35. Conditions: whole-route reservation, saturated demand, each road one resource.

H1 in the manifesto's three parts: all three have readings; the final wording of parts 2 and 3 is the author's, after T27

1	The capacity interval and the start of the network-bound regime K^* Okpo [627, 1,720], K^* 50–60; Yantai [455, 1,252], K^* 25 (whole-route reservation, saturated demand)	established T18–T22; floor from reservation, T27 re-checks
2	At whole-yard volume, where the vehicles needed sit relative to K^*, and the gap to free flow Okpo's own 339 a day: reservation needs 27 (16 h) / 15 (24 h), below K^*; the peak of 678: 117 / 46; 600 a day: 88 / 37	readings in new numbers and main caliber: the author, after T27
3	Whether the dock mouth becomes the yard's bottleneck depends on whether vehicles waiting for the crane block the way to other stops: where they wait, and whether the dock road is used in sections Waiting off the road (reservation), throughput never drops; rules that wait on the road, teleport-free safe release included, drop with crane utilisation, the threshold moving left with fleet size and right with segmented blocking	readings in two-layer wording: the author, after T27
Vehicles-needed criterion Vehicles needed (mean throughput $\geq 99\% \times$ daily volume) $\geq K^*$ means network-bound Density criterion Daily volume / network bound; volume at whole-yard caliber, converted the same way for both yards Conclusive either way Holds: shipyard transport has network-bound capacity. Not: a density criterion bounds where free flow suffices		

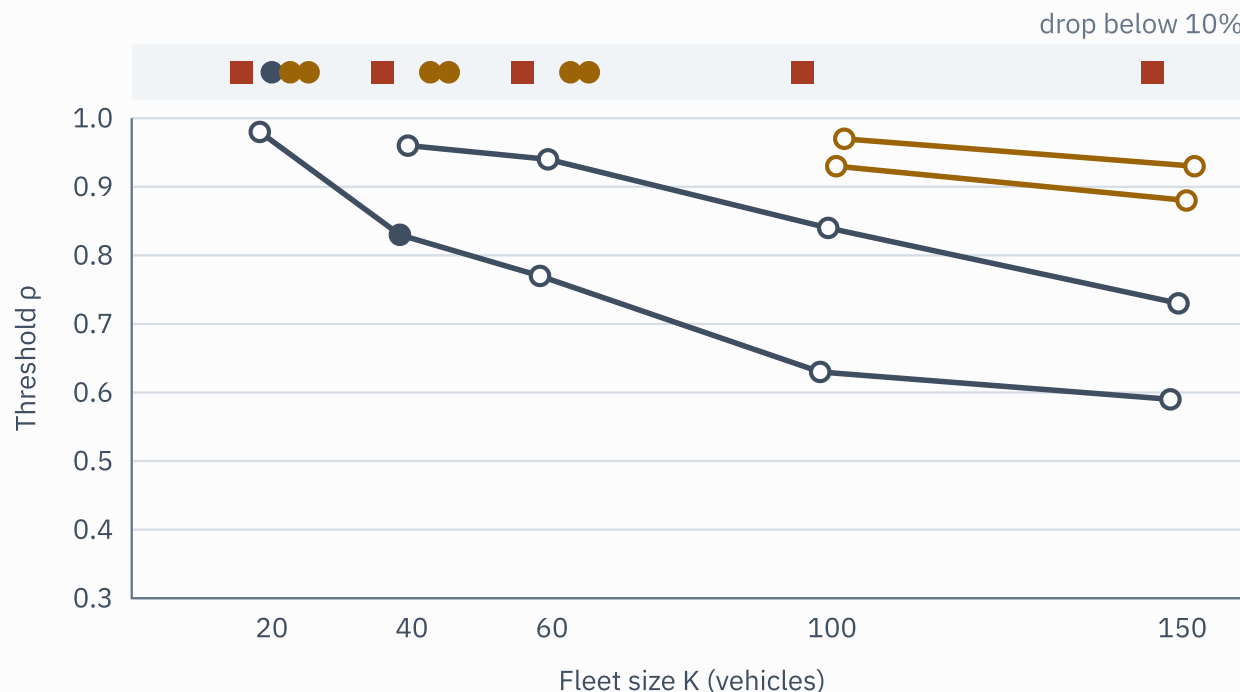
Sources: manifesto v1 §3, H1 (criteria and three parts quoted verbatim); readings in the T18–T22 reports at thesis-core a882742 (T22 §3, §6, §7; suggested wording §15); the three items awaiting the author in 00_总控/reviews/T22.md §5 and tasks.md, row T22 (10 Oct 2026).

Dock mouth (H1 part 3): rules that wait on the road drop, earlier with more vehicles; reservation never drops

Okpo

Yantai

Crane-cadence p sweep, saturated, 8 days



■ Whole-route reservation (both) — Safe release · whole road — Safe release · segmented

— Reference rule (teleport) · whole road — Reference rule (teleport) · segmented

○ hollow = uses an unsteady point

0.98 → 0.59

Safe release, whole road: the threshold moves from 0.98 at $K = 20$ to 0.59 at $K = 150$ (Yantai 0.93 → 0.38); more vehicles, longer queues on the road

0.83 → 0.96

Okpo, $K = 40$, safe release: segmented blocking at the stops moves the threshold right

- **Crane starvation:** at $p = 0.99$ safe release leaves the cranes 12–25 points less used; under reservation Okpo's cranes still reach 96%
- **Teleport contamination:** under the reference rule about 40–50% of truck-borne P6 moves are teleported at least once on the way to the dock, so the dock conclusions rest on reservation and safe release

Sources: thesis-core a882742, docs/T22_report.md §3.1–3.4 and results/t22_report_3287d62c7916/t22_rho_k.csv (batches 54f5e26030e2, 16d6dde8b72d; whole-road reservation and reference rule reused from T21). Threshold = the crane utilisation at which throughput first falls 10% below the lowest- p level (Okpo 0.53, Yantai 0.13), interpolated between neighbouring levels; saturated, 8 days, 10 seeds; Okpo rev 1825, Yantai rev 396.

Whole-yard caliber: Okpo's own 339 a day and peak 678; vehicles needed to the single vehicle, coloured by the H1 test

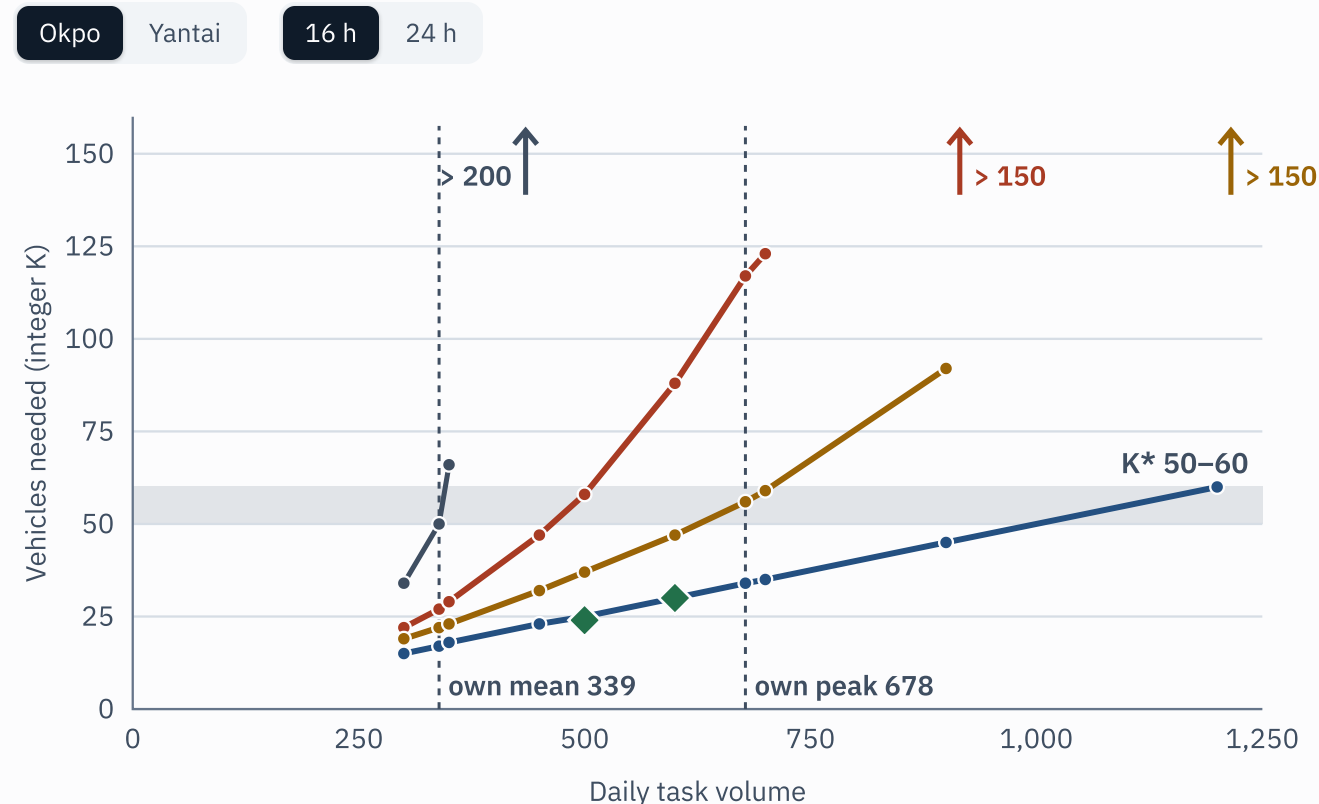
- **Okpo itself:** converted as for Yantai (steel ÷ block weight × moves ÷ working days), 339 a day (294–363 across years); Hanwha Systems' 105,000 moves a year ÷ 300 days = 350, 3% apart; the peak is twice that, 678 / 700
 - **Large Korean yards:** Yim 2008, over 300 a day; Shen 2018, about 600; Hyundai Heavy, about 500 over 24 h. 97 / 194 is a four-transporter subset, kept for comparison
 - **Yantai itself:** 23 / 46 a day by the same method, 1–2 vehicles
- Test:** vehicles needed = the smallest integer K whose 10-seed mean throughput reaches 99% of the daily volume; at or above the top of the K* range it holds, inside the range it is near the start.

Okpo · whole road Okpo · segmented Yantai K* = 50–60								
Tasks a day	16-h working day				24-h working day			
	Free	Resv.	Safe	Ref.	Free	Resv.	Safe	Ref.
300	15	22	34†	19	10	13	13	12
339 own mean	17	27	50†	22	12	15	15	14
350 direct anchor	18	29	66†	23	12	16	16	14
450	23	47	> 200	32	15	22	52†	19
500 Hyundai Heavy scale	25	58	> 200	37	17	27	68†	22
600 Shen et al. scale	30	88	> 150	47	20	37	192†	27
678 own peak	34	117	> 200	56	23	46	> 200	32
700	35	123	> 150	59	24	49†	> 150	33
900	45	> 150	> 150	92	30	95	> 200	47
1,200	60	> 150	> 150	> 150	40	> 150	> 150	73

■ holds (≥ top of K*)
 ■ near the start (inside the K* range)
 ■ does not hold (< bottom of K*)

† a higher fleet falls short again (near the threshold); safe release "> 150" was not scanned further, "> 200" still short after scanning K = 100–200; Yantai's 23 / 46 have 16-h readings only (T20)

The work-zone chart: under a 24-h day, real fleets sit between free flow and whole-route reservation



17 < 24 < 27

About 500 a day over 24 h: the model needs 17 in free flow and 27 under reservation; Hyundai Heavy's 24 sit between

20 < 30 < 37

About 600 a day over 24 h: free flow 20, reservation 37 (reference rule 27); Shen et al.'s about 30 sit between. Over 16 h it takes 30-88, and only free flow matches

Working-day length is a first-order parameter: Hyundai Heavy states 24 h, while Shen et al. and Okpo publish no shifts. Both days are shown; the author settles the main caliber after T27. The comparison takes a real fleet as the vehicles needed (no maintenance or spare margin), so it reads as magnitude and order.

Sources: thesis-core a882742, t22_demand.csv (whole road, integer K) and docs/T22_report.md §6.2; field fleets: Park & Seo 2006 (Hyundai Heavy, 24 vehicles, 24 h, about 500 a day) and Shen et al. 2018 (about 600 a day, about 30 vehicles, shift not stated), both checked in full text by T24. K* is the main-scenario reservation range. With segmented dock roads, reservation's vehicles needed differ from whole roads by at most 1. Conditions: Okpo rev 1825, Yantai rev 396.

The H1 test under reservation: Okpo's mean is not network-bound, its peak and 600 a day are on 16-h days

Volume caliber	16-h working day	24-h working day
Okpo own mean (339 same method / 350 direct anchor) K* 50–60	does not hold reservation 27–29	does not hold reservation 15–16
Okpo own peak (678 / 700, mean × 2) K* 50–60	holds reservation 117–123	lower edge of the start reservation 46–50
Hyundai Heavy scale (500 a day) K* 50–60	near the start reservation 57–58	does not hold reservation 27
Large Korean yard scale (600 a day) K* 50–60	holds reservation 88	does not hold reservation 36–37
Yantai own (23 / 46) K* 25	does not hold reservation 1–2	already below at 16 h
Yantai at Korean scale (450 a day) K* 25	holds reservation 69	does not hold reservation 19

The test

Vehicles needed at or above the top of the K* range: holds; inside it: near the start; below it: does not hold. Each Okpo cell covers both granularities.

Conditions

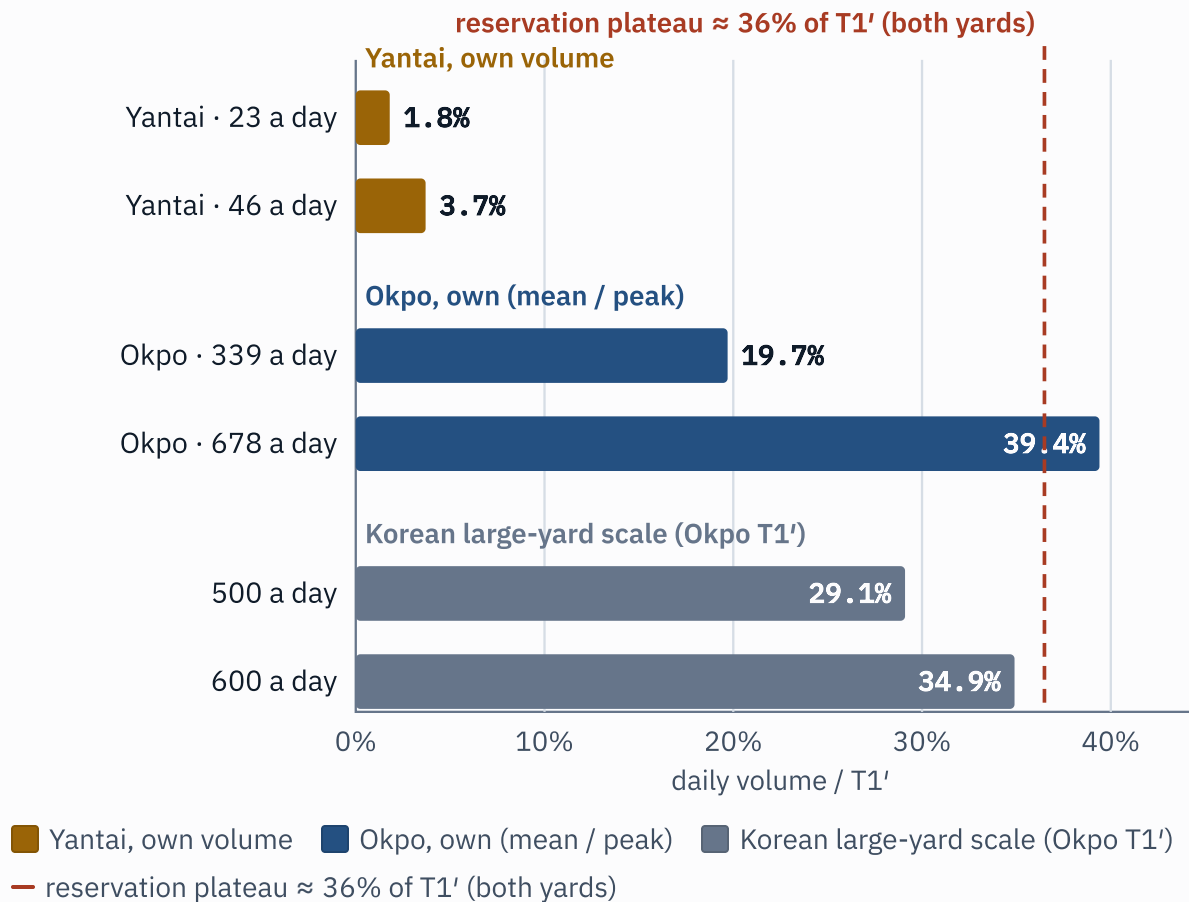
Whether the yard is network-bound depends on the volume caliber (mean or peak), the working day (16 h or 24 h) and the traffic rule; Okpo publishes no shifts, Hyundai Heavy states 24 h.

The author decides after T27

H1 part 2's new numbers, part 3's two-layer wording, and the main caliber (16 h or 24 h, mean or peak). This slide shows both calipers without choosing.

Sources: thesis-core a882742, docs/T22_report.md §7, §15, t22_h1.csv and t22_demand.csv; K* ranges from T18 (Okpo 50 [50, 60]) and T20 (Yantai 25 [25, 25]); vehicles for Yantai's own 23 / 46 from T20 (16 h). Pending decisions: 00_总控/reviews/T22.md §5. Conditions: main-scenario flows; Okpo rev 1825, Yantai rev 396.

Density puts the yards on two sides of H1: Yantai's own volume is 2–4% of T1' , Okpo's 20% (mean) and 39% (peak)



Yantai: the side where free flow suffices

Its own 23–46 a day need 1–2 vehicles, far below $K^* 25$; at a Korean large-yard scale of 450 a day (16 h), reservation would need 69, network-bound too.

Okpo: below the regime at its mean, inside at its peak

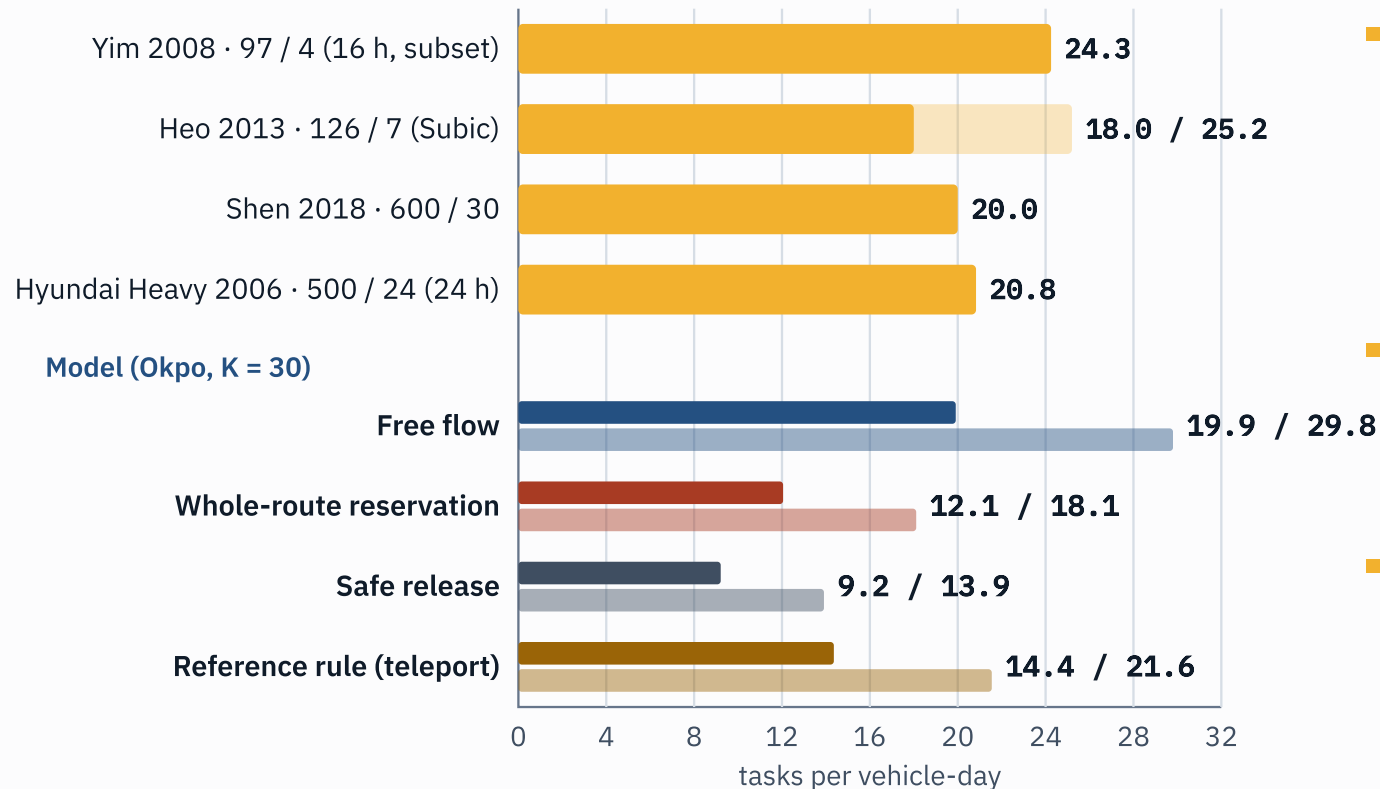
Its own mean of 339 is 20% of T1', with 27 vehicles under reservation on 16-h days; the peak of 678 is above the saturated reservation plateau of 627 and needs 117 on 16-h days (with finite demand idle vehicles stay off the road, the reservation table is emptier, and the volume is still reached).

Yantai's T1' binds at the two doors of building 014 (doors held exclusively while loading). The author's imagery check: vehicles can pass in front, so this granularity is conservative; T25 adds a door-granularity scenario.

Sources: this deck's make_data.py (daily volume $\div$ T1'; T1' is the interval's ceiling, thesis-core a882742, Okpo a0ab16fceb94, Yantai b82aa19b22af); vehicles needed in t22_demand.csv and T22 §6.1 (the emptier reservation table under finite demand: end of §6.1); Yantai's volume in the T20 report §3; binding element and imagery check in 00_总控/tasks.md (9 Oct 2026). Both yards' reservation plateaus are 36% of T1'. The Korean large-yard scale is taken against Okpo's T1', for reference only. Conditions: saturated plateau, each road one resource; Okpo rev 1825, Yantai rev 396.

External anchors (T24, full texts): 18–25 blocks per vehicle-day; normalise for cycle time and utilisation first

Field anchors

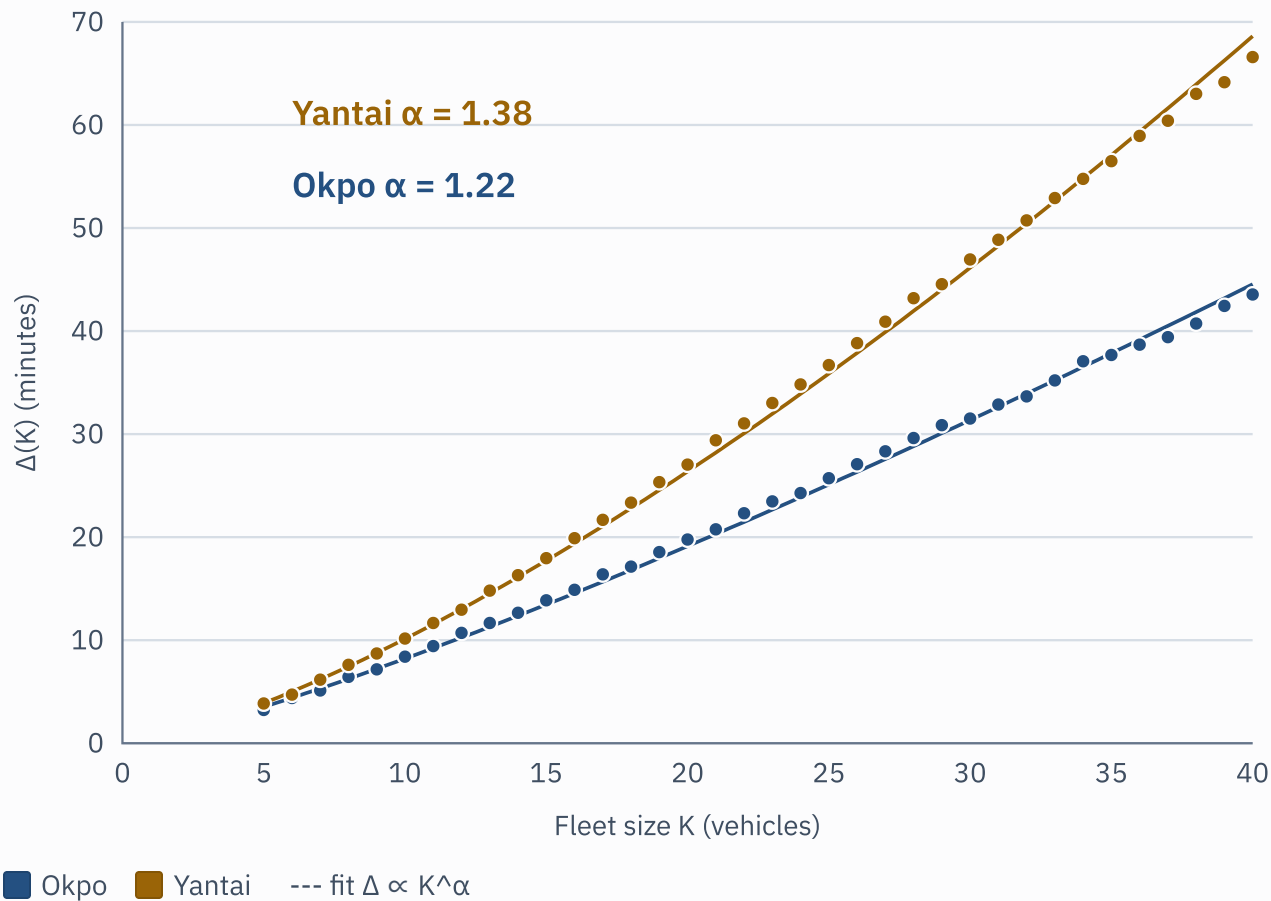


Field anchors 16 h 24 h

- Normalise:** output per vehicle = hours × busy share ÷ cycle. Okpo's model cycle without waiting is 48.3 min (empty 9.1, loading 10, loaded 19.1, unloading 10); Yim's 24.25 needs a cycle ≤ 39.6 min, yet those 4 vehicles were busy only about 27% of the day: the match at about 20 comes from cycle and utilisation cancelling
- Fleets:** Hyundai Heavy, 24 vehicles for about 500 a day over 24 h, against the model's 17 in free flow and 27 under reservation; Shen et al., about 30 for about 600 a day, against 20 and 37. Real fleets sit between the two (work-zone chart)
- Road load (next):** Song & Kang 2009 simulate 102–161 passes a day and up to about 26% use on roads around a stockyard; by the authors' affiliation the yard is inferred to be DSME Okpo; to go into chapter 3's external comparison table against the model's per-road passes

Sources: Yim et al. 2008 (97 blocks / 4 vehicles, 8 a.m. to midnight), Heo et al. 2013 (Subic, 126 / 7; the light bar is 25.2 counting the 5 main vehicles), Shen et al. 2018, Park & Seo 2006 (Hyundai Heavy) and Song & Kang 2009, all checked in full text by T24 (Reference/notes/occupancy_anchors_20261009.md §1.1–1.8); model values from t22_anchor.csv at thesis-core a882742 (Okpo, K = 30, saturated, 16 h and 24 h) and T22 §6.3. Conditions: Okpo rev 1825, each road one resource; the reference rule is for comparison only.

H3: the service-time gap is superlinear ($\alpha = 1.22$ / 1.38), and the vehicle ratio rises with volume



Manifesto criterion: service-time gap $\Delta(K) \propto K^\alpha$ with $\alpha > 1$; the ratio of vehicles needed is read at whole-yard volume.

44 / 67 min

Extra mean service time (dispatch to unloaded) of whole-route reservation over free flow at K = 40: Okpo / Yantai

Tasks a day	16 h: free → reservation	ratio	24 h: free → reservation	ratio
300	15 → 22	1.5	10 → 13	1.3
339	17 → 27	1.6	12 → 15	1.3
450	23 → 47	2.0	15 → 22	1.5
600	30 → 88	2.9	20 → 37	1.9
678	34 → 117	3.4	23 → 46	2.0
900	45 → > 150	> 3.3	30 → 95	3.2

Sources: mean service time (saturated) in curves.csv of each batch at thesis-core a882742, Δ = reservation – free flow; α is the log-log least-squares slope over K = 5–40 (as in experiments/t17_flag.py, line 595). Okpo's 1.22 matches summary.json; fitted over all K (2–150), Yantai gives 1.29 (summary.json). Ratios of vehicles needed from t22_demand.csv (T22 §6.1). Conditions: whole-route reservation (just in time), each road one resource; Okpo rev 1825, Yantai rev 396.

What is established and what is still being settled: T22, T23 and T24 established; T27 running; T25 and T26 queued after it

Established (Okpo rev 1825, Yantai rev 396; whole and segmented roads)

- **Bounds:** T1' checked case by case on 57,917 runs, none above; system T1' removes sampling noise, T1'' holds for any routing rule
- **Capacity interval:** Okpo [627, 1,720] (segmented [630, 1,857]), Yantai [455, 1,252]; erection mix [611, 703] → [662, 1,241]
- **Best deadlock-free rule:** whole-route reservation; safe release is exact and reliable but conservative
- **Readings for all three parts of H1:** K^* ; integer vehicles needed (16 h / 24 h); the dock-mouth mechanism
- **H3:** $\alpha = 1.22$ and $1.38 > 1$; the vehicle ratio rises with volume
- **External anchors (T24):** on 24-h days field fleets sit between free flow and reservation

Still being settled (running or queued, not results)

- **T27 (running):** a less conservative, still reliable deadlock-free rule; re-checks the floor, H2's fall and H5's cost of safety
- **Author's decision (after T27):** H1 part 2's numbers, part 3's two-layer wording, the main caliber
- **T25 (queued):** retrofit simulations and the exchange-rate table, door granularity, block-length curve, the three-resource dock network
- **T26 (queued):** floating docks; finite holding area S and $K_o(S)$
- **Theorems:** draft proofs of T1' and T1'' await the author's rewrite; T5 is being tested in T27

Next: T27 first, then the author's three H1 decisions, then T25 and T26; each step with the figure, table or section it feeds

STEP	CONTENT	OUTPUT (PLACE IN THE THESIS)	STATUS
T27	A less conservative, still reliable deadlock-free rule: Banker plus a step-wise feasibility search (grant if either holds), or claims cut at the next safe holding point (T5); report the share of waits from rejection; step zero: switch to 20_数据/船厂地图 (Okpo rev 1833), close the direction-lock U-turn gap at stops, split reservation waits into crane and road, disambiguate resource names	Chapter 4 rule comparison; chapter 3 interval table, floor column; re-check of H2's fall and H5's cost of safety	running
Author	Decide together: H1 part 2's new numbers, part 3's two-layer wording, the main caliber (16 h or 24 h, mean or peak)	Final wording of H1; chapter 3 H1 test table	after T27
T25	On T27's code and rev 1833: retrofit simulations and the first road–vehicle exchange-rate table; capacity against block length; the three-resource dock network; Yantai's door granularity; road load in the external comparison	Chapter 5 exchange-rate table; the C3 / C4 rule; chapter 3 dock section and external comparison table	queued
T26	Floating-dock scenario (quay-side sink and floating crane); finite holding area S and $K_0(S)$	C2's falling part (H2); chapter 5, what a holding area is worth in vehicles; the H7 case	queued

Sources: 00_总控/tasks.md, rows T25–T27 (10 Oct 2026: the author chose T27 before T25); docs/T22_report.md §5, §10, §15; 00_总控/reviews/T22.md §5.

Risks and mitigations: each has a way through

RISK	MITIGATION
Safe release claims conservatively, so the floor currently uses reservation	T27's less conservative rules plug into the same batches and readings; the floor rises and the interval narrows, and H2's fall is re-checked at the same time
Whether Okpo is network-bound moves with the caliber: not at its mean, yes at its peak and at 600 a day on 16-h days	H1 is conclusive either way: holding at the peak shows network-bound capacity, not holding at the mean bounds where free flow suffices; both calipers are reported and the author sets the main one after T27
Neither Okpo nor Shen et al.'s yard publishes shifts, and the fleet comparison depends on working-day length	On 24-h days real fleets already sit between free flow and reservation (Hyundai Heavy states 24 h); normalise further by cycle time and utilisation (Hanwha Systems' 6 km mean haul as an input), as a column of the external comparison table
The 200 m and 25 m of segmented blocking are scenario settings, not field measurements	Whole and segmented roads are reported side by side; T25 draws capacity against block length, turning one setting into a curve; the author checks passing places at the dock on the imagery
Competition in the field: the closest domestic neighbour is Tao et al. 2017	Tao 2017 read in full: a single-yard simulation counting road capacity in vehicles to compare vehicle types, with no blocking-time formalism, rule comparison, deadlock or bound; it does not narrow the claim and the introduction names it; submit soon as planned

Sources: docs/T22_report.md §10 (risks and strengthening), §6.2, §6.3; T24 note §1.6, §3.1 (Tao 2017 read in full and separated point by point, T24b); 00_总控/reviews/T24.md §5.

Paper one (chapters 2–3: bound, capacity interval, dock mouth): target journals from the school's 2026 directory

When transporters own the road: capacity bounds and operating regimes in shipyard block logistics

Working title; “and dock-mouth instability” may be added if the dock proposition holds

- **Model and bounds:** T1' and the route-free T1'', with width bin-packing and crane rows
- **Capacity interval and work zones:** integer vehicles needed, the K^* range and the density criterion, matched in magnitude to field fleets
- **Dock mouth:** where vehicles wait, how the road is sectioned, how many wait on it
- **The C3 / C4 boundary and the first road–vehicle exchange-rate table (T25)**

B-class first choice; the A-class journal is a stretch, with its condition on the right.

First choice (B)

Transportmetrica A: Transport Science (Class B) · Ocean Engineering (Class B)

Transport science: occupancy model, bounds, rule comparison; naval engineering: two real yards, dock mouth, retrofits

Stretch (A)

Reliability Engineering & System Safety (Class A)

Condition: the dock proposition proved as a stability / resilience result (blocking-induced starvation and recovery)

Alternatives (B)

IEEE Transactions on Automation Science and Engineering (Class B) · Applied Mathematical Modelling (Class B)

T-ASE: capacity and release control in automated logistics; AMM: the LP bound and blocking model

Supplementary (C)

Computers & Industrial Engineering (Class C) · Journal of Ship Production and Design (Class C)

Manufacturing-logistics and shipyard-production readers

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Data and files: results/ at thesis-core commit a882742 (the T18–T23 batches named in each slide's source line; the T22 summary t22_report_3287d62c7916) and the docs/T17–T23 reports; research manifesto v1 (tag manifesto-v1); the T24 note Reference/notes/occupancy_anchors_20261009.md; 00_总控/tasks.md and reviews/T22–T24. Maps traced by the author on public imagery; this briefing contains neither imagery nor network geometry.

What this sub-study adds over the closest work in the field, and where to go next

1

A bound that holds for any rule

T1' writes occupancy as a structural capacity constraint, and T1'' extends it to any routing rule. Roh & Cha 2011 treat interference as a soft term in the objective; in the search, no comparable bound appears in the shipyard literature.

2

Capacity as a measurable interval

Measured on two real yards and two granularities as [best deadlock-free rule, T1'], with the floor at 36% of the ceiling in the main scenario; the erection mix gives examples of the C3 / C4 boundary.

3

Regime criteria that work both ways

Integer vehicles needed, the K^* range and the density criterion sort yards into the two sides and match field fleets in magnitude: on 24-h days real fleets sit between free flow and reservation.

← Thesis briefing

Ceiling: network bound T1' →

Floor: traffic orchestration →

H1 part 3: the dock mouth →

Fleet sizing (MY-B) →

Capacity is not one number but an interval: the traffic rule sets its floor, the network its ceiling.

Thank you.

Sources: Roh & Cha 2011's "interference" as a soft objective term, checked in full text by the literature library (00_总控/tasks.md, 9 Oct 2026); the closest work in the field, Tao et al. 2017, counts road capacity in vehicles to compare vehicle types, with no blocking-time formalism, rule comparison, deadlock or bound (read in full, T24b); search scope: appendix B of the review and T24 (shipyard transport scheduling, AGV zone control, lifelong multi-agent path finding). MY-B's flows P1–P8, block sizes and vehicle parameters: thesis-core T17 report §1.3 and §2.